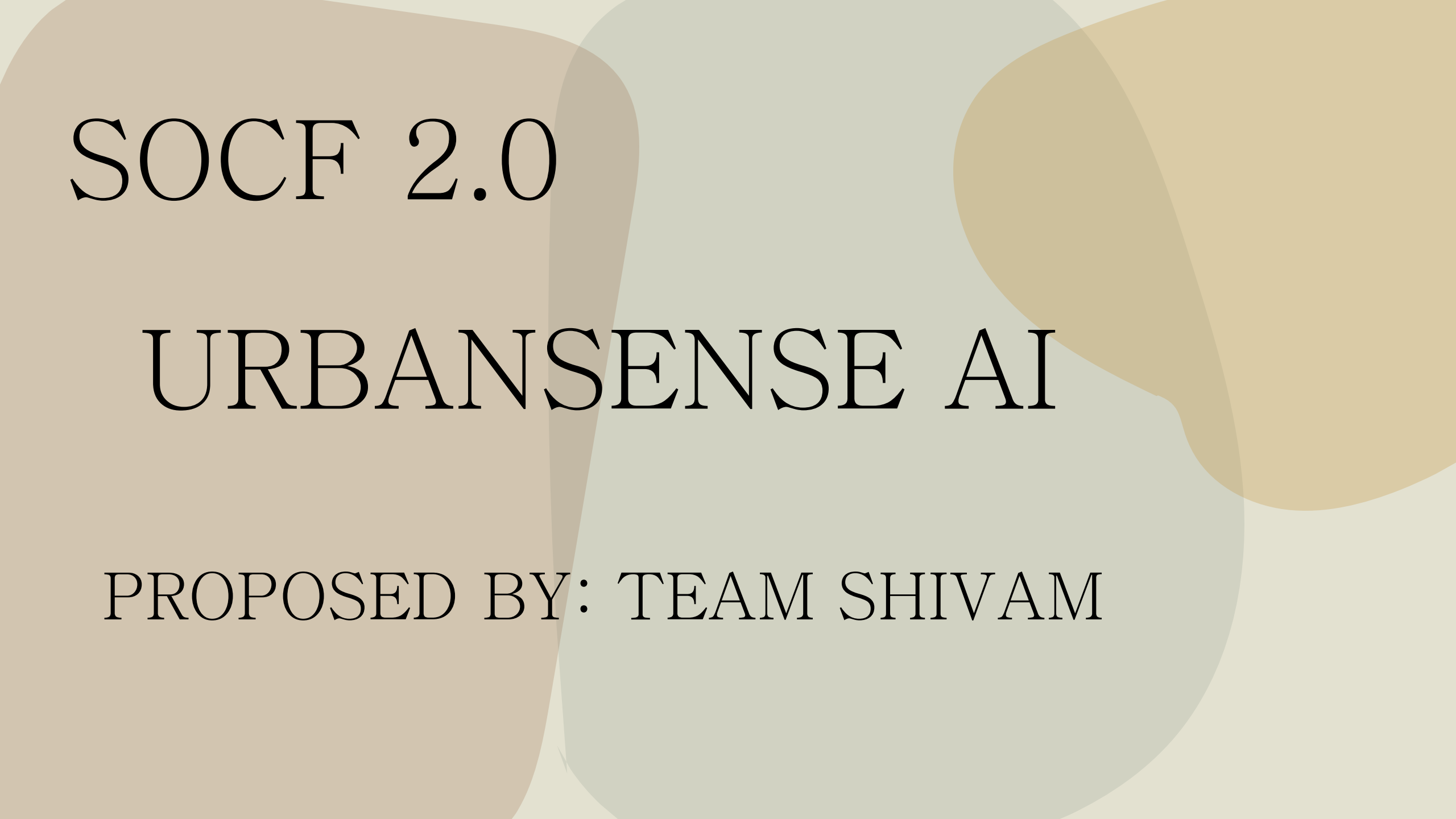




SOCF 2.0

URBANSENSE AI

PROPOSED BY: TEAM SHIVAM

PROBLEM STATEMENT

- Urban areas face challenges in efficiently managing civic complaints due to manual processes, delayed responses, duplicate complaints, and lack of transparency. There is a need for an AI-powered system that streamlines complaint reporting, prioritization, tracking, and resolution to improve urban governance and citizen satisfaction.

PROPOSED SOLUTION

- **Urban Sense AI** is an AI-powered Smart City complaint management platform that helps citizens report civic issues and enables government authorities to manage complaints efficiently. The system automates complaint classification, prioritization, duplicate detection, and department assignment while providing real-time tracking and analytics.

Easy complaint reporting through a mobile app.

AI-based complaint classification

Duplicate and spam complaint detection.

Image validation for authentic complaints.

TECHNICAL STRATEGIES

- **Frontend:** Flutter mobile app for complaint reporting and tracking.
- **Backend:** Fast API for API development and complaint management.
- **AI Module:** Complaint classification, priority prediction, spam & duplicate detection, image validation, and department assignment.
- **Database:** SQLite/PostgreSQL for storing users and complaint data.
- **Maps:** Google Maps API for location-based complaint visualization.
- **Analytics:** Dashboard with complaint statistics and performance insights.
- **Deployment:** Render (Backend), Firebase (Authentication & Image Storage), GitHub (Version Control).

Methodology

- **Complaint Registration** – Citizens submit complaints with description, image, and location through the Flutter app.
- **Data Processing** – The Fast API backend receives and validates the complaint data.
- **AI Analysis** – The AI module classifies the complaint, predicts priority, detects spam/duplicates, validates the image, and assigns the responsible department.
- **Database Storage** – Complaint details and AI results are securely stored in the database.
- **Complaint Management** – Government authorities view, track, and update complaints through the dashboard.
- **Monitoring & Analytics** – Real-time maps, charts, and analytics help monitor complaints and support faster decision-making.
- **Status Updates** – Citizens can track the progress and resolution status of their complaints through the mobile application.

Implementation Process (Diagrams/Flowcharts)

- **Complaint Submission** – Citizen submits complaint with image and location.
- **Backend Processing** – Fast API validates and processes the request.
- **AI Analysis** – Complaint is classified, prioritized, and verified.
- **Data Storage** – Complaint and AI results are stored in the database.
- **Dashboard Display** – Authorities monitor complaints through maps and analytics.
- **Status Tracking** – Citizens receive real-time complaint updates.

FEASIBILITY AND VIABILITY

- **Feasibility**

- Uses proven technologies: **Flutter**, **Fast API**, **Python**, **Firebase**, and **Google Maps API**.
- AI modules can be integrated with the existing backend.
- Scalable architecture suitable for smart city applications.
- Can be deployed on cloud platforms like **Render**.

- **Viability**

- Cost-effective and easy to maintain.
- Reduces manual effort through AI automation.
- Improves complaint resolution time and transparency.
- Can be adopted by municipal corporations and smart city authorities.
- Easily scalable to support multiple cities and increasing users.

Strategies to Overcome Challenges

- **AI-Based Automation:** Automate complaint classification and prioritization to reduce manual effort.
- **Duplicate & Spam Detection:** Filter invalid and repeated complaints to improve efficiency.
- **Image Validation:** Verify uploaded images to ensure complaint authenticity.
- **Scalable Architecture:** Use Flutter and Fast API for a modular and scalable solution.
- **Secure Data Management:** Protect user data with authentication and secure database practices.
- **Real-Time Monitoring:** Use dashboards and maps for faster decision-making and resource allocation.
- **Cloud Deployment:** Deploy on cloud platforms (e.g., Render) for high availability and easy maintenance.

Impact and benefits envisaged

- **Faster Complaint Resolution** through AI-based prioritization and routing.
- **Improved Transparency** by enabling citizens to track complaint status in real time.
- **Reduced Manual Effort** with automated complaint classification and verification.
- **Better Resource Allocation** using analytics and interactive dashboards.
- **Enhanced Citizen Satisfaction** through efficient and reliable public services.
- **Scalable Solution** that can be adopted by multiple smart cities and municipal authorities.

Why may this idea fail ?

- **Inaccurate AI Predictions** may lead to incorrect complaint classification or prioritization.
- **Poor Quality Images or Incomplete Data** can reduce the accuracy of AI analysis.
- **Low User Adoption** if citizens are unaware of or reluctant to use the platform.
- **Lack of Government Support** may delay complaint resolution despite successful reporting.
- **Scalability Challenges** if the system receives a very high number of complaints without adequate infrastructure.
- **Internet Connectivity Issues** may affect real-time complaint submission and tracking.

THANK YOU